School of Computer Science Engineering and Technology

Course- **BTech**
Course Code- **CSET340**

Type- **Specialization Elective**
Course Name- **Advanced Computer Vision and Video Analytics**

Year- **2026**
Date- **02-03-2026 to 06-03-2026**

Semester- EVEN
Batch- **2023-2027 (VI Sem)**

Lab Assignment No. 6

Exp. No.	Name	CO-1	CO-2	CO-3
1.	To perform various Color image Processing and transformations	✓	✓	✓

Objective: Apply various color image processing and transformations on different images on various color images via the use of Python using OpenCV/Pillow/Ski Images Library? Perform the provided task as follows.

Data Set: Download the Zip_Folder containing "Test_Images_lab_6" images with some colored, gray scaled and black and white images.

1. Use the following RGB images named **"lena_image" and "face"**, convert its pixel values to different color model namely **CMY, CMYK, HSV and YCbCr** and produce the output images. **2 Marks**
2. Perform the following Pseudo color image processing techniques namely **"2-slice Intensity Slicing", "8-slice Intensity Slicing", "2-color only coding"** on "medical" image **and "multi-color intensity color coding"** on the **"lena-grey", "map" and "bag"** images. **3 Marks**
3. Perform Intensity to color transformations by feeding gray-level transforms namely log, power-law and piecewise transforms to RGB channel mapping for composite color image generation of the **"lena-grey", "map" and "butterfly"** images. **2 Marks**
4. Perform color transformations namely **negative transform, log transform, power law transform, image smoothening, sharpening and tone correction** on **"Lena_image", "face" and "pyramid"** images. **3 Marks**

Note:- Suggested Platform: Python: Jupyter Notebook/Visual Studio Code/Google Colab.
Mode of Delivery: Face-to-face: Instructor-led discussion and live coding demonstration. Hands-on Practice via Google Colab/VS code/Notebook.
Submission: On LMS within the prescribed time frame, else marks will.